

Generalized Concept Unlearning Guided by AI Feedback for Text-to-Image Diffusion Models

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Abstract

Text-to-image diffusion models have shown remarkable generative capabilities, but they also tend to produce inappropriate or sensitive visual concepts—raising ethical and legal concerns. Existing concept removal techniques mostly rely on keyword-based embedding editing, which makes them fragile to paraphrased, misspelled, or indirect prompts and often leads to unintended suppression of semantically related concepts. We propose Generalized Concept Unlearning (GCU), a novel reinforcement learning framework that performs output-grounded semantic concept erasure guided by pretrained AI feedback (e.g., vision-language models). Instead of keyword-based model editing, GCU optimizes a multi-objective reward to unlearn the target concept while maintaining image fidelity and prompt alignment. Our approach is unified and extensible: it supports a wide range of concept types by modifying only the reward configuration. Furthermore, we employ prompt variation strategies to achieve robustness against adversarial and paraphrased prompts, and preserve non-target concepts and context. GCU also enables fine-grained control over the outcome of erased regions, allowing users to specify whether removed concepts should leave empty space or be replaced with semantically related alternatives. Our source code is in the supplementary.

Introduction

Text-to-image diffusion models have recently demonstrated remarkable capabilities in generating high-quality and diverse images from natural language prompts. These models have been rapidly adopted in various domains such as advertising, art, and content creation (Rombach et al. 2022; Ramesh et al. 2021; Saharia et al. 2022; Nichol et al. 2021). However, because they are trained on large-scale web-crawled datasets, they often learn and reproduce inappropriate or sensitive visual concepts such as pornography, violence, discrimination, and content that infringes upon copyright or portrait rights. This raises serious ethical, social, and legal concerns, and poses a critical challenge to the responsible deployment of generative AI technologies (Kumari et al. 2023; Gandikota et al. 2023, 2024; Zhang et al. 2024; Huang et al. 2024).

Early efforts to mitigate these risks have relied on post-hoc filtering (Bedapudi 2020; Rando et al. 2022) or inference-time guidance (Schramowski et al. 2023). While simple to implement, these methods do not alter the model’s

internal representations and can be easily bypassed, especially in open checkpoint environments. As a more proactive solution, recent works have proposed model editing approaches that modify internal representations associated with prompt keywords to directly remove specific concepts from diffusion models (Kumari et al. 2023; Gandikota et al. 2023, 2024; Zhang et al. 2024; Huang et al. 2024).

However, existing concept editing methods suffer from three major limitations. (1) They primarily rely on manipulating embeddings associated with specific keywords, making them vulnerable to paraphrased, misspelled, or indirect prompts. That is, while the target concept may be suppressed when using the original keyword, it often reappears when semantically similar expressions are used. (2) Unintended concept damage may occur, where semantically related but non-target concepts are also affected. For example, removing “Grumpy Cat” may unintentionally degrade the model’s ability to generate general cat images. (3) Concept removal can degrade prompt-image alignment, removing not only the target concept but also the context of the image. For instance, when given a prompt such as ‘Celebrity X being interviewed by reporters,’ the model may not only omit the target celebrity, but also fail to depict the surrounding scene, including the reporters and the interview context.

To address these limitations, we propose the Generalized Concept Unlearning (GCU) framework. Instead of keyword-based representation editing, GCU adopts an output-based reinforcement learning approach that evaluates generated images using pretrained AI models (e.g., CLIP (Radford et al. 2021), LLaVA (Liu et al. 2023), LPIPS (Zhang et al. 2018)). The reward function is designed not only to penalize the presence of the target concept but also to encourage the preservation of non-target semantics, ensuring minimal unintended concept damage. By adjusting only the reward configuration without altering the model architecture, GCU can be flexibly applied to five representative concept types; objects, artistic styles, NSFW content, celebrities, and specific instances. It achieves targeted concept unlearning while maintaining overall image quality and text-image consistency, even under diverse prompt rephrasing. For instance, when unlearning a specific instance, related non-target concepts can be included in the prompt to prevent unintended concept damage during training.

Through extensive experiments, we demonstrate that our



Figure 1: The proposed GCU framework effectively removes various types of visual concepts—including objects, NSFW content, artistic styles, celebrities, and instances—within a single unified approach. From Stable Diffusion 1.4 (top), GCU (bottom) selectively erases the target concept at semantic-level, while preserving non-target concepts and contextual information.

method achieves more robust and semantically precise concept removal than prior editing approaches, while preserving the model’s ability to generate non-target content. This highlights the practical utility and generalizability of the proposed unlearning framework.

Related Works

Removing Sensitive Contents. Recent text-to-image diffusion models (Rombach et al. 2022; Saharia et al. 2022; Ramesh et al. 2021; Nichol et al. 2021) have demonstrated impressive generative capabilities, but also raised serious concerns about misuse, copyright infringement, and the production of NSFW or otherwise harmful content. In response, early mitigation strategies were developed to restrict inappropriate outputs without modifying model weights. These include post-hoc filtering (Bedapudi 2020; Rando et al. 2022), which applies external classifiers or heuristics after image generation, and inference-time guidance (Schramowski et al. 2023), which steers the sampling trajectory away from unsafe regions by modifying the generation dynamics. For example, Safe Latent Diffusion (SLD) (Schramowski et al. 2023) injects negative concept guidance into the latent space using a pretrained safety classifier, effectively discouraging unsafe content during inference. While lightweight and easy to deploy, these methods operate externally to the model and are often bypassable by users with access to model checkpoints or generation code.

Model Editing for Concept Erasure. To overcome the limitations of external mitigation, several works have proposed to directly edit or fine-tune diffusion model weights to permanently remove specific concepts (Gandikota et al. 2023, 2024; Zhang et al. 2024; Kumari et al. 2023; Huang et al. 2024). Ablating (Kumari et al. 2023) fine-tunes the model so that the generative distribution of the target concept is aligned with that of a predefined anchor concept,

effectively preventing the model from producing the target concept while preserving closely related content. Erasing Concepts from Diffusion Models (ESD) (Gandikota et al. 2023) uses a negative guidance fine-tuning method that permanently erases a visual concept by updating the U-Net’s weights under the supervision of a frozen teacher model. Universal Concept Erasure (UCE) (Gandikota et al. 2024) takes a training-free approach by computing a closed-form approach that directly modifies the cross-attention projection matrices (W_k and W_v) to steer specific concepts toward edited representations, while preserving others to minimize interference. Forget-Me-Not (FMN) (Zhang et al. 2024) adopts attention rerouting: it minimizes cross-attention weight of the target concept in the diffusion U-Net, progressively disregarding the concept while largely preserving other content. Receler (Huang et al. 2024) learns a small set of additional weights via LoRA (Hu et al. 2022) to reliably disable generation of the target concept. Its training process incorporates concept-localized regularization, which confines the model edits to the target concept’s representation, and an adversarial prompt augmentation scheme to ensure robust erasure even against paraphrased or tricky prompts.

Preliminaries

Denosing Diffusion Models. Diffusion models are generative models that learn the data distribution by modeling a gradual denoising process and maximizing a variational bound on the data likelihood. Starting from random Gaussian noise x_T , the model predicts noise ϵ_t at each time step t to progressively reconstruct the original sample x_0 . This process is formulated as a Markov chain with the joint probability:

$$p_\theta(x_{T:0}) = p(x_T) \prod_{t=T}^1 p_\theta(x_{t-1} | x_t) \quad (1)$$

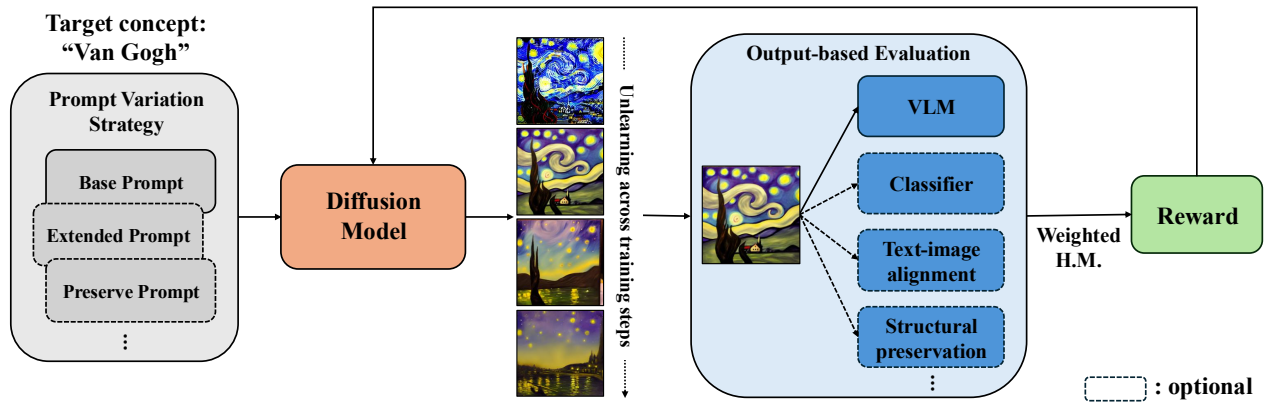


Figure 2: Overview of GCU. The proposed framework employs a prompt variation strategy to expose diverse expressions of the target concept, and an output-based evaluation to optimize the model via RL using composite reward.

Reward-Driven Fine-Tuning of Diffusion Models.

The conventional training objective of diffusion models—maximizing a variational bound on the data likelihood—has been successful in learning high-fidelity image distributions. However, they often fail to align with human preferences such as aesthetics or prompt-image alignment. To address this, recent works have proposed reward-driven fine-tuning approaches that optimize the model using reinforcement learning based on output-level feedback. Methods like DDPO (Black et al. 2023), DPOK (Fan et al. 2023), and B2-DiffuRL (Hu et al. 2025) treat the denoising process as a Markov Decision Process, where each step is an action guided by rewards computed from the final image—e.g., aesthetic quality score or CLIP similarity. These approaches enable controllable, preference-aligned generation without modifying the model architecture.

Motivation

Recent studies on concept erasure in diffusion models (Gandikota et al. 2023, 2024; Kumari et al. 2023; Zhang et al. 2024; Huang et al. 2024) primarily focus on modifying internal representations associated with specific keyword prompts. These methods offer fast and localized concept removal by editing the cross-attention layers or the text encoder. However, such keyword-based representation editing tends to perform only superficial erasure: the concept may disappear under exact keywords but still persist when expressed through paraphrased or indirect prompts. This limitation reflects a lack of semantic robustness in existing methods.

To address this issue, Receler (Huang et al. 2024) introduced adversarial prompt learning, which augments fixed tokens with trainable soft prompts to better generalize across paraphrased and obfuscated expressions. While this broadens the scope of erasure, it also expands the embedding-level erasure boundary, leading to unintended concept damage. For instance, removing the concept of “Grumpy Cat” may degrade the model’s ability to generate cat images in general; similarly, removing ‘Elon Musk’ not only eliminates distinctive facial features but can also cause the model to

fail in generating any human figure at all

These challenges highlight the need for a semantically grounded and generalizable concept-unlearning framework. Rather than modifying internal representations, we propose to perform unlearning by directly evaluating and optimizing the model’s outputs. Our output-grounded approach enables robust semantic unlearning that is resilient to paraphrased, indirect, or misspelled prompts; it precisely removes the target concept while preserving non-targeted concepts and contextual contents.

Methodology

We propose Generalized Concept Unlearning (GCU), a novel framework for semantically removing specific visual concepts from text-to-image diffusion models. Unlike prior approaches that edit internal representations tied to specific keywords, GCU evaluates the semantic content of generated outputs and leverages reinforcement learning guided by reward signals derived from pretrained evaluators. This enables GCU to go beyond fixed keyword expressions and remain robust to paraphrased, misspelled, or indirect prompts.

While this reward-driven optimization is inherently more computationally expensive than keyword-based editing methods, it enables semantically grounded concept unlearning under realistic, diverse, and unpredictable prompt variations. More importantly, GCU offers controllability over the outcome of concept removal. For instance, when erasing an object, the model can be guided—via reward configuration—to either leave the region blank or replace it with a semantically related alternative. Furthermore, the reward design allows us to preserve image fidelity and prompt-image alignment during the unlearning process.

Since GCU operates at the output level, it can be applied to a wide range of concept types without modifying the model architecture or inference pipeline—only by reconfiguring the reward function. In this work, we demonstrate the generality of GCU across five categories: objects, NSFW content, artistic styles, celebrities, and specific instances.

Reinforcement Learning for Concept Erasure. In the context of concept erasure, our goal is to reduce the like-

likelihood that a diffusion model generates visual content associated with a given target concept. That is, we aim to minimize the probability density that the model assigns to samples exhibiting the concept. Formally, let $p_\theta(x_0 \mid c)$ denote the conditional distribution over images x_0 given a prompt c , and let $r(x_0, c)$ denote the reward function that measures the degree of target concept removal in the generated image. Our objective is to optimize the model parameters θ such that:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{x_0 \sim p_\theta(\cdot \mid c)} [r(x_0, c_{\text{target}})]. \quad (2)$$

However, the conditional distribution $p_\theta(x_0 \mid c)$ is intractable and non-differentiable, as it marginalizes over a high-dimensional latent trajectory and involves stochastic sampling steps. Therefore, following the approach of DDPO (Black et al. 2023), we formulate the denoising process as a multi-step Markov Decision Process (MDP), where each denoising step is treated as a state transition.

Under this formulation, we apply reinforcement learning to optimize the model via trajectory-level rewards. Specifically, our objective can be rewritten in the PPO-style policy gradient form:

$$\nabla_{\theta} \mathcal{L}(\theta) = \mathbb{E} \left[\sum_{t=1}^T \min \left(r_t(\theta) \hat{A}_t, \text{clip}_{\epsilon} (r_t(\theta)) \hat{A}_t \right) \right], \quad (3)$$

where $r_t(\theta)$ is an importance sampling ratio ($\frac{\pi_{\theta}(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$), $\hat{A}_t$ is an advantage estimate, and clip_{ϵ} is a clipping function that constrains the policy ratio within $[1 - \epsilon, 1 + \epsilon]$. This formulation enables optimization from non-differentiable, image-level rewards and allows the concept suppression signal to be propagated throughout the generation process.

Reward Function Design. The reward function plays a central role in the concept unlearning framework by providing quantitative feedback that guides the model to unlearn visual representations associated with a target concept. We design a composite reward function that integrates the outputs of multiple pretrained evaluation models, each capable of robustly detecting the target concept and capturing aspects such as image quality and semantic alignment with the prompt.

We combine the following pretrained AI evaluators to construct the reward function, and the framework allows flexible integration of new or improved models as they become available.

- **(1) VLM-based concept detection:** We use pretrained vision-language models (e.g., LLaVA (Liu et al. 2023)) to evaluate the presence of a target concept in a generated image x_0 with an evaluation prompt. The full prompt template is provided in the supplementary material.
- **(2) Classifier-based recognition:** For concepts that can be explicitly classified (e.g., objects, nudity), we employ image classifiers such as ResNet-50 (He et al. 2016) and NudeNet (Bedapudi 2020) directly use their predicted inclusion probabilities as reward signals.
- **(3) Text-to-Image alignment:** To ensure that the model maintains semantic consistency, We evaluate the alignment between the image and a modified prompt without

the target concept using CLIP (Hessel et al. 2021). Details of the prompt modification strategy are provided in the supplementary material.

- **(4) Structural preservation:** To ensure that the model does not degrade or hallucinate unnatural structures while erasing the target concept, we compute structural consistency using LPIPS (Zhang et al. 2018).

Because each component operates on a different scale and direction, we normalize all scores to the range $[0, 1]$, where higher is better. The final reward is computed as the weighted harmonic mean of the four normalized scores:

$$r(x_0, c) = \text{Weighted HM} \left(R_{\text{VLM}}, R_{\text{Cls}}, R_{\text{Align}}, R_{\text{struct}} \right) \quad (4)$$

The weighted harmonic mean formulation emphasizes balance while allowing flexible control over the relative importance of concept suppression, semantic alignment, and image quality. This multi-reward structure enables the model not only to suppress the target concept effectively, but also to maintain semantic coherence and perceptual realism in its outputs.

Prompt Variation Strategy. Most existing concept removal methods depend on fixed keyword-based prompts, which makes them fragile to paraphrased, indirect, or misspelled expressions. In contrast, our output-grounded reward design enables training with a diverse set of prompt types, thereby improving both robustness to prompt variations and controllability over the removal process.

For fine-grained learning and robust unlearning, we suggest three prompt categories. This design is extensible and can accommodate additional categories as required for different concept types or application domains:

- **(1) Base prompts:** Simple expressions that directly mention the target concept (e.g., "a photo of Elon Musk").
- **(2) Extended prompts:** Compositional prompts that contain the target concept along with additional contextual elements (e.g., "Elon Musk standing next to a red car"). This category also includes adversarial variants—longer prompts with paraphrases, typos, or indirect references to the target concept (e.g., "A tech entrepreneur at Gigafactory", "Elo Msk near a tree")—designed to challenge the model’s robustness.
- **(3) Preserve prompts:** Prompts that explicitly mention non-target concepts to be retained (e.g., "Bill Gates"), inspired by the preservation strategy in UCE (Gandikota et al. 2024).

To generate diverse and semantically rich prompt variations with minimal manual effort, we use ChatGPT-4o (OpenAI 2025). This lightweight yet flexible prompt generation strategy significantly enhances the robustness and consistency of concept unlearning in our framework.

Experiments

Experiment Setup. To evaluate the effectiveness of our General Concept Unlearning (GCU) framework, we conduct experiments across five representative categories of

Method	Base Prompts				Extended Prompts			
	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$
SD 1.4	84.8 ± 11.7	77.9 ± 2.89	N/A	N/A	69.1 ± 19.1	69.0 ± 2.12	0.3102 ± 0.01	0.3102 ± 0.00
ESD	1.5 ± 2.97	70.5 ± 3.52	N/A	N/A	7.7 ± 8.35	37.1 ± 2.15	0.2640 ± 0.02	0.2969 ± 0.00
UCE	0.7 ± 1.47	73.0 ± 3.14	N/A	N/A	2.8 ± 3.84	39.0 ± 2.55	<u>0.2698 ± 0.00</u>	<u>0.3000 ± 0.01</u>
Receler	<u>0.1 ± 0.20</u>	70.0 ± 3.98	N/A	N/A	<u>2.5 ± 5.04</u>	49.5 ± 2.82	0.2415 ± 0.02	0.2986 ± 0.00
Ours	0.0 ± 0.0	70.3 ± 4.84	N/A	N/A	2.4 ± 2.96	<u>46.5 ± 4.62</u>	0.2713 ± 0.02	0.3012 ± 0.00

Table 1: **Quantitative evaluation of object removal on 10 ImageNet classes.** Acc_e and Acc_r are measured using ResNet-50.

visual concepts: **Object**, **NSFW (Nudity)**, **Artistic Style**, **Celebrity** and **Instance**.

We used Stable Diffusion v1.4 (Rombach et al. 2022) as the backbone text-to-image diffusion model for all experiments. To enable efficient fine-tuning, we applied Low-Rank Adaptation (LoRA; Hu et al. 2022) to the UNet backbone of Stable Diffusion.

For reward functions, we employed LLaVA-7B (Liu et al. 2023) for semantic detection. Additionally, we incorporated ResNet-50 (He et al. 2016) and NudeNet (Bedapudi 2020) as classifier-based evaluators, CLIP (Hessel et al. 2021) for image-text alignment, and LPIPS (Zhang et al. 2018) to quantify structural preservation. We note that specific configurations vary slightly by task to better align with the nature of the target concept and the availability of evaluators. Detailed reward compositions are described in supplementary materials.

Prompt Generation. Both training and evaluation prompts were constructed using ChatGPT-4o (OpenAI 2025). An exception is made for the NSFW (Nudity) task, where we use the I2P dataset provided by (Schramowski et al. 2023) as the source of evaluation prompts instead of generating new ones. Details of prompt construction for each task are provided in the supplementary materials.

Evaluation Metric. We use four quantitative metrics to evaluate both the effectiveness of concept erasure and the fidelity of preserving non-target content:

- **(1) $\text{Acc}_e^{(\downarrow)}$** : Measures how frequently the erased concept still appears in images generated from target prompts.
- **(2) $\text{Acc}_r^{(\uparrow)}$** : Measures how well non-target concepts are preserved in images generated from non-target prompts.
- **(3) $\text{Align}_e^{(\uparrow)}$** : Measures the CLIP similarity between the generated image and the modified prompt in which the target concept is replaced, indicating how well the model preserves the remaining semantic content after unlearning.
- **(4) $\text{Align}_r^{(\uparrow)}$** : Measures the CLIP similarity between non-target prompts and generated images to assess whether prompt-image alignment is preserved after unlearning.

For tasks where a reliable concept classifier is available (Object, NSFW), Acc_e and Acc_r are computed using classifier accuracy. In contrast, for more abstract categories (Artistic Style, Celebrity), we follow the approach of (Kumari

et al. 2023) and use CLIP accuracy (Hessel et al. 2021). It compares the CLIP similarity between the generated image and a concept (e.g., Van Gogh) against an anchor concept (e.g., Artist), and count the prediction as correct if the concept score is higher. All anchor concepts used in this setup are consistent with the destination concepts defined in UCE (Gandikota et al. 2024). We highlight the best-performing results in bold and the second-best in underline for clarity.

Results

Object Erasure. To evaluate object-level concept removal, we conduct experiments on 10 Imagenette (Howard 2019) class labels, using the same target classes as prior works such as ESD and UCE to ensure comparability. For each class, we fine-tune the model to unlearn the corresponding concept and evaluate the erasure performance. We evaluate the models using two types of prompts: (1) base prompts, which directly mention the target concept (e.g., “a photo of {target}”), and (2) extended prompts, which include prompts with additional context and their adversarial variations. For the extended prompt setting, we generate new 100 unseen prompts per class that were not used during training to assess generalization performance.

Table 1 reports the average performance across the 10 classes. Compared to prior methods, our approach achieves a superior trade-off between concept removal accuracy (Acc_e) and non-target preservation (Acc_r), while also maintaining text-image alignment (Align_e & Align_r). This demonstrates that our method effectively removes the target object while preserving contextual information and fidelity with respect to non-target prompts.

Note that alignment scores were not measured (N/A) for explicit prompts, as simple descriptions like “a photo of {target}” lack meaningful contextual information for alignment assessment. Qualitative examples of object removal are provided in the supplementary materials.

NSFW (Nudity) Erasure. We further evaluate GCU’s ability to remove NSFW content, focusing specifically on nudity. From the 4,703 prompts in the I2P benchmark (Schramowski et al. 2023), we select 854 prompts where the nudity percentage is greater than 0% according to the original dataset annotations. Since it is difficult to define clear remaining concepts in the context of nudity, we restrict our evaluation to erasure accuracy only.

Class	SD 1.4	ESD	UCE	Receler	Ours
Female Breast	418	137	38	46	5
Armpits	324	132	34	71	4
Belly	208	78	22	30	4
Feet	86	34	11	31	2
Buttocks	80	31	8	19	6
Male Breast	57	19	4	6	0
Female Genitalia	48	8	0	6	2
Male Genitalia	27	13	8	11	5
Total	1248	452	125	220	28
Percentage	0.00%	-63.8%	<u>-90.0%</u>	-82.4%	-97.8%

Table 2: **Quantitative evaluation of NSFW (nudity) removal.** Each value represents the number of detected occurrences across 8 nudity-related categories, measured using NudeNet on images generated from 854 I2P prompts.

Method	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$
SD 1.4	0.780	0.740	0.3087	0.2936
ESD	0.390	<u>0.633</u>	0.2838	0.2795
UCE	0.500	<u>0.558</u>	0.3145	0.2938
Receler	0.220	0.570	0.2704	0.2766
Ours	<u>0.360</u>	0.650	<u>0.2956</u>	<u>0.2830</u>

Table 3: **Quantitative evaluation of artistic style removal on *Van Gogh*.** Acc_e and Acc_r are measured using CLIP accuracy between the target/non-target concept and an anchor concept.

Table 2 summarizes the number of detected occurrences across eight nudity-related classes for each method. Lower values indicate better erasure performance. Our method achieves the lowest total count, effectively suppressing NSFW content across all categories. These results demonstrate that GCU performs the most effective nudity removal among all compared baselines. Qualitative results are provided in the supplementary materials.

Artistic Style Erasure. We then evaluate GCU’s capability to remove artistic styles. Specifically, we target *Van Gogh* as the concept to be erased. For evaluation, we generate 100 unseen prompts related to *Van Gogh* and 900 prompts covering the other 9 artists (e.g., *Picasso*, *Munch*).

Quantitative results in Table 3 show that Receler achieves the lowest Acc_e , indicating strong erasure. However, its Acc_r , Align_e , and Align_r scores are substantially lower than other methods, suggesting that over-suppression has occurred and On the other hand, UCE obtains the highest alignment scores, but performs poorly in both Acc_e and Acc_r , suggesting inconsistent concept control. In contrast, our method achieves a well-balanced performance, significantly reducing the presence of the target style (Acc_e) while best preserving non-target concepts (Acc_r) and maintaining reasonable alignment, demonstrating a balanced and robust approach to style erasure. Qualitative results are provided in the supplementary materials.

Method	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$
SD 1.4	0.920	0.683	0.3087	0.3191
ESD	0.540	0.658	0.2838	0.2923
UCE	0.470	0.517	0.3145	0.3186
Receler	<u>0.440</u>	<u>0.662</u>	0.2704	0.2860
Ours	0.340	0.663	<u>0.2920</u>	<u>0.2923</u>

Table 4: **Quantitative evaluation of celebrity removal on *Elon Musk*.** Acc_e and Acc_r are measured using CLIP accuracy between the target/non-target concept and an anchor concept.



Figure 3: **Qualitative result of celebrity removal on *Elon Musk*.** Our method removes the target concept with minimal destruction.

Celebrity Erasure. Following a similar configuration to the artistic style setting, we evaluate the performance of GCU on the celebrity erasure task. We target *Elon Musk* as the concept to be removed. A total of 1,000 prompts are used for evaluation, including 100 unseen prompts related to *Elon Musk*, and 900 prompts associated with nine other celebrities(e.g., *Bill Gates*, *Angelina Jolie*).

As shown in Table 4, our method demonstrates strong and balanced performance across all metrics. Unlike artistic style removal, celebrity erasure not only requires confirming whether the concept has been removed, but also calls for visual inspection of how the removal is reflected in the generated image. This makes qualitative analysis essential, as quantitative metrics alone may not capture the perceptual quality of the outcome. For example, Receler achieves the second-lowest Acc_e after ours, indicating strong erasure capability. However, the qualitative comparison in Figure 3 reveals a critical distinction: GCU selectively removes only the facial features associated with *Elon Musk*, while Receler tends to degrade the entire human-like structure. These results demonstrate that GCU not only effectively removes the target concept but also faithfully preserves non-target information, confirming its robustness and precision in the celebrity erasure task.

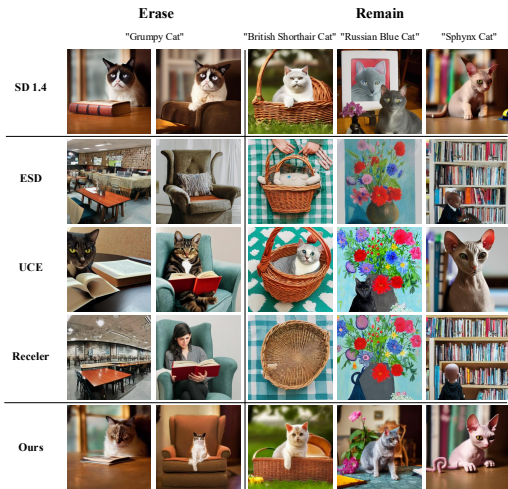


Figure 4: **Qualitative results of instance-level removal on *Grumpy Cat*.** Our method effectively removes *Grumpy Cat* while preserving high-level concept (*Cat*) as well as other cats (e.g., *British Shorthair Cat*)

Instance Erasure. Finally, we conduct a qualitative evaluation on instance-level concept removal. The goal in this setting is to remove a specific instance (e.g., *Grumpy Cat*) that belongs to a broader category (e.g., *cat*), while preserving the semantics of the higher-level concept. Figure 4 shows that GCU successfully removes the *Grumpy Cat* instance from the generated images while maintaining the overall concept of *cat*. Other subcategories, such as *British Shorthair Cat* and *Russian Blue Cat*, also remain well-preserved. In contrast, prior methods such as ESD and Receler often fail to generate coherent *cat*-like images after the removal of *Grumpy Cat*, indicating that the higher-level concept is inadvertently suppressed along with the target instance. These results demonstrate that GCU’s output-based approach enables fine-grained, semantic-level concept erasure, effectively eliminating the target instance while preserving semantically related but non-target subcategories.

Ablation Study. To assess the impact of reward design and prompt strategies, we conduct ablation experiments on the *cassette player* concept under the extended prompt setting. Table 5 presents results for different reward combinations. Using only the VLM-based reward yields an Acc_e of 0.0, suggesting perfect erasure. However, Acc_r , Align_e , and Align_r drop significantly, revealing that the model fails to preserve non-target content and contextual information. This indicates a form of model corruption, where overall generation quality is compromised. Adding a classifier-based reward mitigates this, slightly increasing Acc_e while significantly improving non-target and alignment metrics. Further including CLIP and LPIPS leads to additional gains in preservation metrics, without compromising erasure. These results highlight the importance of balanced reward design for effective and semantically faithful unlearning.

Table 6 shows the effect of different prompt strategies. The addition of (2) extended prompts and (3) preservation

Reward Functions				Extended Prompts			
VLM	Cls	Align	Struct	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$
✓				0.0	31.9	0.2054	0.2844
✓	✓			3.5	42.3	0.2620	0.2952
✓	✓	✓		<u>2.5</u>	<u>49.1</u>	<u>0.2866</u>	<u>0.3010</u>
✓	✓	✓	✓	<u>2.5</u>	50.1	0.2979	0.3018

Table 5: **Ablation study on the effect of reward functions.** *Cls*, *Align*, and *Struc* refer to the classifier reward, alignment reward, and structural preservation reward, respectively.

Prompt Type			Extended Prompts			
(1)	(2)	(3)	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$
✓			3.0	40.8	0.2830	0.2980
✓	✓		2.0	<u>45.9</u>	0.2756	<u>0.3030</u>
✓	✓	✓	<u>2.5</u>	50.1	0.2979	0.3018

Table 6: **Ablation study on the effect of different prompts.** The prompt type indices (1)–(3) correspond to those introduced in the *Prompt Variation Strategy* section

prompts leads to meaningful improvements in Acc_r , Align_e , and Align_r . This demonstrates that including richer contextual cues and explicit preservation targets in the prompts helps retain non-target concepts and improves semantic consistency.

Limitation

GCU demonstrates robust unlearning performance across diverse concept types and prompt variations. However, due to its RL-based nature, it requires more computational resources compared to prior methods. Additionally, designing effective reward functions may demand domain-specific knowledge depending on the concept category. As with prior methods, our approach does not guarantee complete removal of the target concept. Moreover, there is no theoretical guarantee regarding the preservation of non-target concepts. Future work should explore improving training efficiency, enhancing robustness to issues such as reward hacking, and ensuring high image quality and diversity after unlearning.

Conclusion

We proposed Generalized Concept Unlearning (GCU), an output-based RL framework for semantically removing undesirable concepts from text-to-image diffusion models. By leveraging AI feedback as reward signals, GCU enables precise and context-aware unlearning across diverse concept types while minimizing degradation of non-target generation quality. Extensive experiments demonstrate GCU’s strong generalization, semantic precision, and preservation of image fidelity. Our work offers a practical and extensible solution toward safer and more controllable generative models.

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A Implementation Details

A.1 Prompt Details

Training Prompt Generation Each task employs a tailored set of prompts selected from the three types (Base, Extended, Preserving) to balance robustness and consistency. The prompt composition for each task is as follows:

- **Object:** 100 erasing prompts (20 base, 80 extended) and 100 preserving prompts.
- **NSFW:** We generated and used 200 prompts based on inappropriate concepts such as 'hate', 'harassment', 'violence', 'suffering', 'humiliation', 'harm', 'suicide', 'sexual', 'nudity', 'bodily fluids', and 'blood', which are also used in ESD (Gandikota et al. 2023) and UCE (Gandikota et al. 2024). Preserve prompts are excluded due to the ambiguity of defining a positively preserved NSFW counterpart.
- **Artistic Style:** 50 erasing prompts (5 base, 45 extended).
- **Celebrity:** 100 erasing prompts (4 base, 96 extended) and 100 preserving prompts.
- **Instance:** 200 erasing prompts and 200 preserving prompts.

Evaluation Prompt Generation Each evaluation task is conducted using carefully curated prompt sets to assess the model's robustness, alignment, and content preservation capabilities. The composition and generation method for each task are as follows:

- **Object:** We conduct two types of prompt-based evaluation: base and extended. Base prompt is a simple prompt that directly mention the target concept ("an image of {target_concept}"), following the evaluation setup of prior works (Gandikota et al. 2023, 2024). We evaluate on 10 ImageNette (Howard 2019) classes, generating 200 images per class with different random seeds, resulting in a total of 2,000 images for evaluation. Extended prompts include 200 ChatGPT-4o-generated (OpenAI 2025) prompts per class, designed to evaluate indirect or varied expressions. Imagenette consists of following classes: *cassette player, chain saw, church, English springer, French horn, garbage truck, gas pump, golf ball, parachute, tench*.
- **NSFW:** Out of the 4,703 prompts in the I2P (Inappropriate Image Prompts) benchmark, we used 854 prompts whose nudity percentage is greater than zero. These prompts are real user-generated text-to-image prompts that are disproportionately likely to produce inappropriate images.
- **Artistic Style:** For each artist, 100 prompts—composed of base and extended types—were generated using the ChatGPT-4o (OpenAI 2025). We selected 10 artists commonly supported by Stable Diffusion 1.4, resulting in 1,000 prompts in total. The artists include: *Salvador Dalí, Frida Kahlo, Claude Monet, Edvard Munch, Pablo Picasso, Jackson Pollock, Vincent van Gogh, Johannes Vermeer, Leonardo da Vinci, Andy Warhol*.
- **Celebrity:** For each celebrity, 100 prompts—comprising base and extended types—were generated using the ChatGPT-4o (OpenAI 2025). We selected 10 celebrities frequently recognized by Stable Diffusion 1.4, yielding a total of 1,000 evaluation prompts. The selected celebrities are: *Bill Gates, Elon Musk, Adam Driver, Adriana Lima, Amber Heard, Amy Adams, Andrew Garfield, Angelina Jolie, Anjelica Huston, Anna Kendrick*.

VLM Prompt To measure VLM score, the following prompt was given with the generated image: "Answer a single number between 0 and 10, evaluating how much the given image reflects a {target_concept}, where 0 means no relation, and 10 means perfect reflection of {target_concept}. On the next line, briefly explain the reason for your score."

CLIP prompt To measure prompt-image alignment, we compute the CLIP score between the generated image and a modified prompt where the target concept is replaced with an anchor concept. For example, "{target} on the table" → "object on the table", "Starry night by {target}" → "Starry night by an artist", "photo of {target} jogging" → "photo of a person jogging".

If the original prompt does not explicitly mention the target concept, it is used as-is without modification.

A.2 Task-specific Reward Function Design

In our experiments, we generally define the reward function as a combination of four components: **VLM-based concept detection**, **classifier-based recognition**, **text-image alignment**, and **structural preservation**. Specifically, we use LLaVA-7B for VLM-based evaluation, task-specific classifiers for classifier-based evaluation, CLIP for prompt alignment, and LPIPS for structural consistency.

For tasks where a reliable classifier does not exist (Artistic Style, Celebrity, and Instance), the classifier reward is omitted. In the case of Artistic Style, the style is a global visual property, and maintaining structural similarity may still evoke the erased artist's work. Therefore, we exclude the structural preservation reward for this task. However, if the goal is to preserve structural layout while only removing local stylistic cues (e.g., brush strokes), the LPIPS component can be reintroduced.

In summary, the following task-specific reward configurations are used:

- **Object:** LLaVA + ResNet + CLIP + LPIPS
- **NSFW (Nudity):** LLaVA + NudeNet + CLIP + LPIPS
- **Artistic Style:** LLaVA + CLIP
- **Celebrity:** LLaVA + CLIP + LPIPS
- **Instance:** LLaVA + CLIP + LPIPS

A.3 Resource Details

GCU experiments were conducted on a single 48GB A6000. Unlike DDPO (Black et al. 2023), LLaVA inference was conducted on the same GPU as that of StableDiffusion v1.4, taking approximately 1 hour per 12 training iteration.

A.4 GCU Implementation

One training iteration consists of sampling and training process. We collect 16 samples per iteration during sampling. During training we use batch of size 2, accumulate gradients over 2 steps, and perform 4 gradient updates per iteration. Gradients are always accumulated across all denoising timesteps for a single sample, consistent with the trajectory-level optimization paradigm of DDPO. All the experiments were conducted for 200 iterations, and the best model was chosen for evaluation.

A.5 Baseline Implementation

For all baseline methods (Gandikota et al. 2023, 2024; Huang et al. 2024), we follow the official training settings provided in their respective repositories without modification. Specifically, ESD (Gandikota et al. 2023) and Receler (Huang et al. 2024) are trained using only the target concept keyword, while UCE (Gandikota et al. 2024) is trained with both erasing and preserving concept keywords.

Extended prompts could not be directly applied to the baselines. This is because extended prompts contain contextual information beyond the target concept itself. Under the existing mechanisms of the baselines, such contextual content is either entirely removed (Gandikota et al. 2023; Huang et al. 2024) or replaced with a predefined anchor concept (Gandikota et al. 2024). Both approaches can severely degrade model performance, as removing or replacing rich contextual cues disrupts prompt-image alignment and may lead to unintended content loss.

A.6 Full Hyperparameters

Section	Parameter	Value
Sampling	Denoising steps (T)	50
	η	1.0
	Guidance scale	5.0
	Samples per iteration	16
Optimization	Optimizer	AdamW
	Learning rate	3×10^{-4}
	β_1	0.9
	β_2	0.999
	ϵ	10^{-8}
	Weight decay	10^{-4}
	Gradient clip norm	1.0
Training	Batch size	2
	Gradient accumulation steps	2
	Gradient updates per iteration	4
	Clip range	10^{-4}
LoRA	Rank	4
	α	1.0
	Dropout	0.0

Table 7: Full hyperparameters used in our experiments.

B Qualitative Results

This section presents qualitative results that could not be included in the main paper due to space constraints. We provide visual comparisons for three representative concept categories discussed in the main text: *Object*, *Nudity*, and *Celebrity*.

Figure 5 shows example generations from our model after removing target concepts across all 10 ImageNette classes. These results highlight the model’s ability to selectively removal various objects with minimal artifacts or degradation in non-target regions. Figure 6 and Figure 7 illustrate side-by-side comparisons for the removal of *nudity* and the *Van Gogh* artistic style,

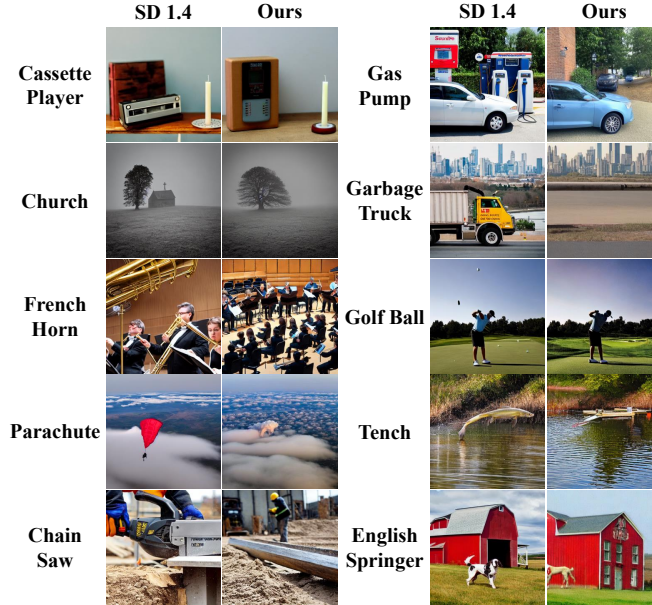


Figure 5: **Qualitative results of object-level concept erasure.** Our method selectively removal various objects with minimal artifacts or degradation in non-target regions.

respectively. We compare our method against ESD, UCE, and Receler. As shown, our method achieves the best trade-off between effectively removing the target style and preserving the remaining non-target styles and overall visual coherence.

C Additional Experimental Results

C.1 Baselines with Adversarial Prompts

As discussed in Section A.5, the other baselines are not inherently designed to leverage extended prompts. Nevertheless, to examine whether these baselines could benefit from exposure to diverse adversarial expressions (e.g., paraphrasing, typos), we conducted an additional experiment.

All adversarial prompts used in our training were derived from extended prompts, containing richer contextual information. To make them compatible with other baselines, we extracted all adversarial expressions (paraphrases and typos) from our training set for each class and used them for fine-tuning. This evaluation was performed on five classes from the Object task—*golf ball*, *parachute*, *garbage truck*, *chainsaw*, and *church*—selected as those with the lowest Acc_e in ESD and UCE. Since Receler (Huang et al. 2024) already employs an adversarial prompt learning mechanism, we restrict this comparison to ESD (Gandikota et al. 2023) and UCE (Gandikota et al. 2024).

The results are presented in Table 8. *Vanilla ESD* refers to the original ESD trained solely on base prompt containing the target concept, while *Vanilla UCE* denotes the original UCE trained with both target and preserving concepts. *ESD (adv.)* and *UCE (adv.)* extend these setups by additionally incorporating adversarial expressions from our extended prompt variations into the training data.

The results indicate that incorporating adversarial expressions leads to performance degradation for both ESD and UCE. For ESD, all four metrics (Acc_e , Acc_r , Align_e , Align_r) decrease noticeably, reflecting diminished erasure effectiveness, non-target preservation, and alignment. For UCE, Acc_e and Acc_r improves slightly, but Align_e drops substantially, suggesting that the model tends to over-suppress, removing not only the target concept but also contextual information that should have been preserved. Qualitative results on training adversarial prompts are shown in Figure 8. This shows that, unlike our approach—which evaluates outputs directly and can robustly adapt to adversarial prompt variations—embedding-editing-based baselines such as ESD and UCE struggle to effectively accommodate such variations, resulting in degraded performance.

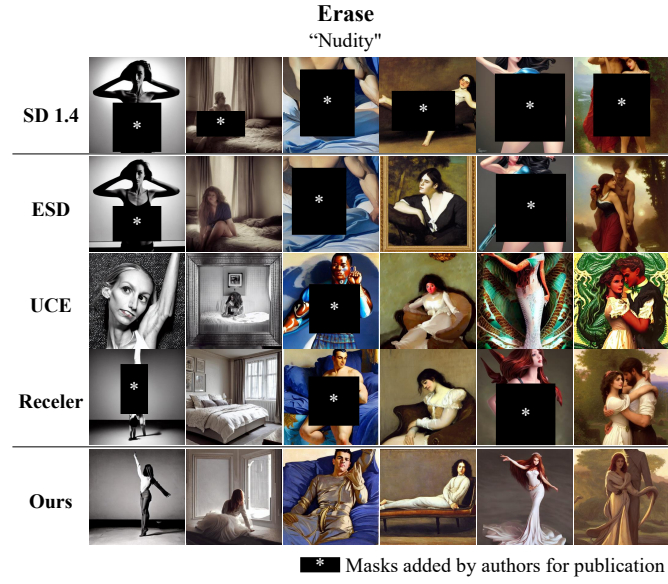


Figure 6: **Qualitative result of NSFW concept removal.** Our method achieves the best trade-off between effectively removing the target style and preserving the remaining non-target styles and overall visual coherence.

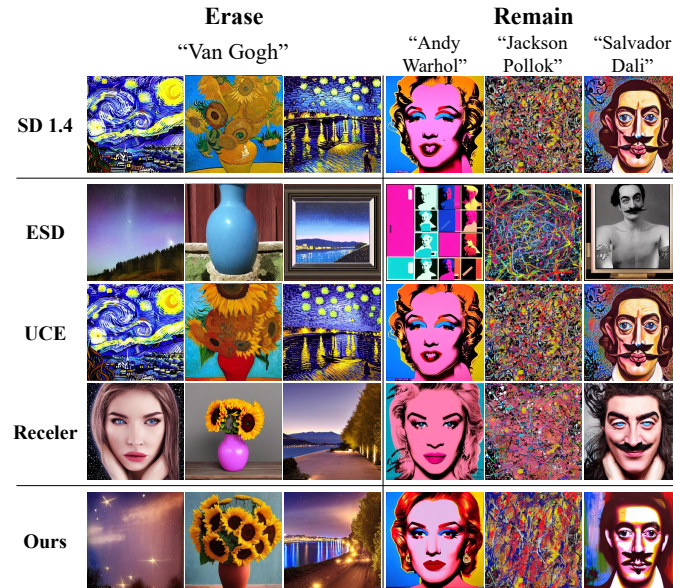


Figure 7: **Qualitative result of artistic style removal.** Our method achieves the best trade-off between effectively removing the target style and preserving the remaining non-target styles and overall visual coherence.

Method	$\text{Acc}_e^{(\downarrow)}$	$\text{Acc}_r^{(\uparrow)}$	$\text{Align}_e^{(\uparrow)}$	$\text{Align}_r^{(\uparrow)}$
Vanilla ESD	10.4 ± 5.68	35.2 ± 1.27	0.2651 ± 0.01	0.2972 ± 0.00
ESD (adv.)	12.2 ± 8.50	31.8 ± 0.85	0.2583 ± 0.01	0.2909 ± 0.00
Vanilla UCE	2.4 ± 0.89	40.9 ± 1.11	0.2739 ± 0.01	0.3064 ± 0.00
UCE (adv.)	1.0 ± 1.00	45.4 ± 2.31	0.2169 ± 0.02	0.2893 ± 0.01
Ours	3.2 ± 4.44	42.7 ± 3.40	0.2564 ± 0.03	0.2998 ± 0.00

Table 8: Quantitative Result on effect of training ESD and UCE with adversarial expressions.



Figure 8: ESD and UCE trained with adversarial expression mostly fails to remain contextual information given in prompt.